

Objectives
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Recall
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Predicate
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Quantifiers
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Negation of Quantification
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AMAT 19

Finite Mathematics

Propositional Function and Quantifiers



Objectives

1. Define a predicate in logical terms;
2. Define universal and existential quantifiers; and
3. Negate universal and existential quantifiers.

Proposition or not?

1. She is beautiful.
2. The lions in Manila Zoo are strong.
3. x is greater than 3.

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1. She is beautiful.
2. The lions in Manila Zoo are strong.
3. x is greater than 3.

All statements are not propositions.

Subject and Predicate

Subject – focus of the sentence

Predicate – refers to anything about the subject

1. She is beautiful.

Subject:

Predicate:

Subject and Predicate

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Predicate – refers to anything about the subject

1. She is beautiful.

Subject: **She**

Predicate: **is beautiful**

2. The lions in Manila Zoo are strong.

Subject:

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Subject and Predicate

Subject – focus of the sentence

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2. The lions in Manila Zoo are strong.

Subject: **The lions in Manila Zoo**

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3. x is greater than 3.

Subject:

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Predicate – refers to anything about the subject

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Subject: **She**

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2. The lions in Manila Zoo are strong.

Subject: **The lions in Manila Zoo**

Predicate: **are strong**

3. x is greater than 3.

Subject: **x**

Predicate: **is greater than 3**

Propositional function/Predicate

Propositional function/Predicate:

A **propositional function/predicate** is a statement that contains variables (predicate variables), and they may be true or false depending on the values of these variables.

When the variables in a propositional function are assigned values, the resulting statement becomes a proposition with a certain truth value.

We introduce a (functional) symbol for the predicate, and put the subject as an argument (to the functional symbol): $P(x)$.

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e.g. x is greater than 3.

Variable: x

Propositional function/predicate: **is greater than 3**

Symbol: Let $P(x)$ denote the statement " $x > 3$ ".

Truth values of propositional functions

1) Let $P(x)$ denote " $x > 3$ ".

What is the truth value of $P(3)$?

Truth values of propositional functions

1) Let $P(x)$ denote " $x > 3$ ".

What is the truth value of $P(3)$? **FALSE**

What is the truth value of $P(25)$?

Truth values of propositional functions

1) Let $P(x)$ denote " $x > 3$ ".

What is the truth value of $P(3)$? **FALSE**

What is the truth value of $P(25)$? **TRUE**

2) Let $A(x)$ denote "Student with student number x belongs to batch 2018."

What is the truth value of $A(2007 - 69833)$?

Truth values of propositional functions

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What is the truth value of $P(3)$? **FALSE**

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2) Let $A(x)$ denote "Student with student number x belongs to batch 2018."

What is the truth value of $A(2007 - 69833)$? **FALSE**

2) Let $Q(x, y, z)$ denote the statement " $x^2 + y^2 = z^2$ ".

What is the truth value of $Q(3, 4, 5)$?

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Example: Consider the predicate $P(x) = \text{"}x^2 \text{ is less than } x\text{"}$.

Then the domain of x could be for example the set $\mathbb{Z}$ of all integers. It could alternatively be the set $\mathbb{R}$ of real numbers.

Whether instantiations of a predicate are true or false may depend on the domain considered.

Quantifiers

Recall: A predicate becomes a proposition when we assign it with fixed values.

However, another way of creating a predicate into a proposition is to **quantify** it.

That is, the predicate is true (or false) for **all** possible values in the universe of discourse or for **some** value(s) in the domain.

1. universal quantifier
2. existential quantifier

Universal quantification

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Example: Here is a formal way to say that for all values that a predicate variable x can take in a domain D , the predicate is true:

$\forall x \in D, P(x)$ (is true).

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Example: Here is a formal way to say that for all values that a predicate variable x can take in a domain D , the predicate is true:

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Example: The square of any real number is nonnegative.

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Example: The square of any real number is nonnegative.

$\forall x \in \mathbb{R}, x^2 \geq 0$, where x denotes “number”, the domain $\mathbb{R}$ is “real number”, predicate $P(x) = x^2 \geq 0$ is “nonnegative”.

Universal quantification

Truth value

“ $\forall x \in D, P(x)$ is true” iff “ $P(x)$ is true for every x in D ”

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All BSAM batch 2019 students of UP Mindanao are lefthanded.

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6. true or false?

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Truth value: What if the domain is empty?

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$\forall x, P(x)$ will still be true since there is no value in the domain for which $P(x)$ will be false.

Universal quantification and Conjunction

Assume that domain D consists of $x_1, x_2, \dots, x_n$,

$\forall x, P(x)$ is the same as the conjunction

$$P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)$$

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“ $2 \cdot 0 = 0 + 0$, and $2 \cdot 1 = 1 + 1$, and $2 \cdot 2 = 2 + 2$, etc.”,
rephrase the statement using universal quantification.

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For all whole numbers x , $2 \cdot x = x + x$.

$$(\forall x \in \mathbb{W}, 2 \cdot x = x + x.)$$

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Example:

There exists a BSAM batch 2020 student of UP Mindanao who lives outside Davao City.

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1. true or false?

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$\exists x, P(x)$ will be false since there is no value in the domain for which $P(x)$ will be true.

Existential quantification and Disjunction

Given a domain D containing $x_1, x_2, \dots, x_n$,

$\exists x, P(x)$ is the same as the disjunction $P(x_1) \vee P(x_2) \vee \dots \vee P(x_n)$

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e.g. Given the statement

“ $0 \cdot 0 = 25$, or $1 \cdot 1 = 25$, or $2 \cdot 2 = 25$, or $3 \cdot 3 = 25$, and so on.”,

rephrase the statement using existential quantification.

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For some whole number x , $x \cdot x = 25$.

$(\exists x \in \mathbb{W}, x \cdot x = 25.)$

Negation of Quantification

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<i>Quantifier</i> (<i>True when</i>)	<i>Negation</i> (<i>False when</i>)
$\forall x, P(x)$	$\sim (\forall x, P(x)) \Leftrightarrow \exists x, \sim P(x)$

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<i>Quantifier</i> (<i>True when</i>)	<i>Negation</i> (<i>False when</i>)
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$\exists x, P(x)$	$\sim (\exists x, P(x)) \Leftrightarrow \forall x, \sim P(x)$

Negation of Quantification

Exercise: Negate the following.

1. All animals are mammals.
2. All BSAM batch 2020 students of UP Mindanao are lefthanded.
3. Some animals are black.
4. There exists a BSAM batch 2019 student of UP Mindanao who lives outside Davao City.

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Answer to Exercise: Negate the following.

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Negation of Quantification

Answer to Exercise: Negate the following.

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3. Some animals are black.
4. There exists a BSAM batch 2019 student of UP Mindanao who lives outside Davao City.

Negation of Quantification

Answer to Exercise: Negate the following.

1. All animals are mammals. **Some animals are not mammals.**
OR **Not all animals are mammals.**
2. All BSAM batch 2020 students of UP Mindanao are lefthanded.
3. Some animals are black.
4. There exists a BSAM batch 2019 student of UP Mindanao who lives outside Davao City.

Negation of Quantification

Answer to Exercise: Negate the following.

1. All animals are mammals. **Some animals are not mammals.**
OR **Not all animals are mammals.**
2. All BSAM batch 2020 students of UP Mindanao are lefthanded. **Some BSAM batch 2020 students of UP Mindanao are not lefthanded.** OR **Not all BSAM batch 2020 students of UP Mindanao are lefthanded.**
3. Some animals are black.
4. There exists a BSAM batch 2019 student of UP Mindanao who lives outside Davao City.

Negation of Quantification

Answer to Exercise: Negate the following.

1. All animals are mammals. **Some animals are not mammals.**
OR **Not all animals are mammals.**
2. All BSAM batch 2020 students of UP Mindanao are lefthanded. **Some BSAM batch 2020 students of UP Mindanao are not lefthanded.** OR **Not all BSAM batch 2020 students of UP Mindanao are lefthanded.**
3. Some animals are black. **All animals are not black.** OR **No animals are black.**
4. There exists a BSAM batch 2019 student of UP Mindanao who lives outside Davao City.

Negation of Quantification

Answer to Exercise: Negate the following.

1. All animals are mammals. **Some animals are not mammals.**
OR **Not all animals are mammals.**
2. All BSAM batch 2020 students of UP Mindanao are lefthanded. **Some BSAM batch 2020 students of UP Mindanao are not lefthanded.** OR **Not all BSAM batch 2020 students of UP Mindanao are lefthanded.**
3. Some animals are black. **All animals are not black.** OR **No animals are black.**
4. There exists a BSAM batch 2019 student of UP Mindanao who lives outside Davao City. **All BSAM batch 2019 student of UP Mindanao do not live outside Davao City.** OR **No BSAM batch 2019 student of UP Mindanao live outside Davao City.**